Subhransu S. Bhattacharjee

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AI PhD candidate at ANU and first-author CVPR/ECCV researcher developing generative world models for 3D scene understanding, uncertainty-aware robotic perception, and planning. Industry research experience spans Microsoft RAG evaluation, Optiver market-microstructure ML, and JPMorgan Chase financial NLP; targeting Applied Scientist, Research Scientist, and Quantitative Research roles.

Education

Doctor of Philosophy — **Artificial Intelligence**
School of Computing, Australian National University, Australia — *Apr 2023 – Mar 2027 (expected)*
- **Thesis topic:** *Into the Unknown: Tackling Spatio-semantic Uncertainty using Generative Posteriors*
- **Supervisors:** Dr. Rahul Shome, Dr. Dylan Campbell, and Prof. Stephen Gould

Bachelor of Engineering — **Mechatronics**
School of Engineering, Australian National University, Australia — *Jul 2018 – Dec 2022*
- **Major:** Mechatronic Systems Engineering, *First Class Honours*
- **Minors:** Mathematics and Electronic Communication Systems; summer school at the London School of Economics
- **Certifications:** Game Theory, Stanford; Machine Learning Production, DeepLearning.AI; Project Management, Google
- **Thesis project:** Whiplash Gradient Descent Dynamics

Industry Research Experience

Microsoft Corporation — **Jun 2026 – Jul 2026**
Applied Scientist PhD Research Intern, LLMs, RAG, Agentic AI
- Built calibrated LLM-as-judge and RAG evaluation frameworks for groundedness under noisy retrieval, providing reusable diagnostics for metric calibration, evaluator selection, and failure analysis.

Optiver APAC — **Nov 2024 – Feb 2025**
Quantitative Research Intern, Machine Learning, Market Microstructure
- Built a real-time ML research pipeline for Korean market microstructure, unifying feature engineering, monitoring, and walk-forward backtesting for faster alpha strategy evaluation.

JPMorgan Chase & Co. — **Dec 2021 – Feb 2022**
Quantitative Research Intern, NLP, Information Retrieval, Risk
- Delivered a transformer-based question-answering system over large financial document collections, turning indexing, retrieval, and answer extraction into client-facing components for risk research.

Academic Research Experience

Research School of Management, Australian National University — **Sep 2023 – Sep 2024**
Graduate Research Assistant — FinTech & AI, PI: Dr Priya Muthukannan
- Developed AI-adoption assessment frameworks for an industry-linked Commonwealth Bank of Australia project, translating open-banking and dynamic-capabilities analysis into strategy recommendations.

CIICADA Lab, Australian National University — **Dec 2022 – Feb 2023**
Summer Research Scholar — Control & Optimisation, Supervisor: Prof Ian Petersen
- Advanced feedback-based optimisation analysis for systems without closed-form solutions, contributing to publications at ASCC 2022 and ANZCC 2021.

School of Computing, Australian National University — **Mar 2022 – Jun 2022**
Undergraduate Research Scholar — Geometry of Data, Supervisor: Prof Richard Hartley
- Tested invertibility of differentiable neural mappings with dense positional encoding, achieving a 72% hit rate under an RMSE criterion.

Selected Publications

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *Believing is Seeing: Unobserved Object Detection using Generative Models. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. Project page.
Introduces unobserved object detection with generative priors that infer occluded and out-of-frame objects across 2D, 2.5D, and 3D scenes.

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *FlatLands: Generative Floormap Completion From a Single Egocentric View. European Conference on Computer Vision (ECCV)*, 2026. Springer. Paper.
Releases FlatLands: 270,575 observations from 17,656 real metric indoor scenes across six datasets.
Compares 11 methods, showing stochastic generators improve completion fidelity and capture layout uncertainty.

Subhransu S. Bhattacharjee, Hao Lu, Dylan Campbell & Rahul Shome: *MatterDoor: Sampling Zero-shot Spatio-semantic Priors using Generative Models. arXiv:2510.11014*, 2025. Under review; RSS 2026 FM4RoboPlan workshop presentation, Sydney. Preprint.
Samples environment priors from partial observations using pretrained generative models.
Produces RGB-D point-cloud hypotheses for occupancy, target semantics, and configuration-space planning.

Subhransu S. Bhattacharjee & Ian Petersen: *Analysis of the Whiplash Gradient Descent Dynamics. Asian Journal of Control* (Special Issue), Wiley, 2023. DOI: 10.1002/asjc.3153.
A feedback-based inertial gradient method with adaptive damping achieves provable polynomial and exponential convergence rates that extend classical optimisation theory to high-dimensional settings.

Subhransu S. Bhattacharjee & Ian Petersen: *A Closed Loop Gradient Descent Algorithm Applied to Rosenbrock's Function. Australian & New Zealand Control Conference (ANZCC)*, 2021. DOI: 10.1109/ANZCC53563.2021.9628258.
Introduces adaptive closed-loop damping for inertial gradient descent on a stiff non-convex optimisation benchmark.

Achievements

2026 Google Research award with ANU Robotics for developing world models for task and motion planning.
2026 Invited to present the FlatLands paper at the Hugging Face, PyTorch, and Red Hat Meetup, Bengaluru.
2025 Invited Research Talk at the Dyson Robotics Laboratory, Imperial College London
2025 Vice-Chancellor's Travel Grant, Australian National University, for CVPR 2025 travel.
2025 Invited attendee, Citadel & Citadel Securities PhD Summit, London.
2024 Optiver PhD Quant Lab Program (selected participant).
2023 ANU International University Research Scholarship (4 years) and Higher Degree by Research Merit Stipend.
2022 Highly Recommended Paper at the *Asian Control Conference*; Best Student Reviewer Award.
2021 High Commendation Award for undergraduate student paper at the *Australia & New Zealand Control Conference*.

Teaching

Senior/Head Tutor, Australian National University: Introduction to Machine Learning; Building Cyber-Physical Systems; Responsible Leadership; Power Systems & Control Theory. Delivered ML mathematics, optimisation, robotics, microprocessor, and NAO robot labs.

Service

Reviewing: ECCV 2026; NeurIPS 2026; ICML 2026; CVPR 2026; ICRA 2025–2026; IEEE Robotics and Automation Letters (RA-L) 2026; AAAI 2025; IROS 2025; Asian Journal of Control (2023); American Control Conference (2022); ANZCC (2021–2022).

Technical Skills

Programming & ML Frameworks: Python; PyTorch; SQL; C; MATLAB; Bash; NumPy; pandas; SciPy; scikit-learn; PySpark; LaTeX.
CV, Robotics & Generative AI: OpenCV; Open3D; PyTorch3D; TorchVision; ROS2; COLMAP; 3D reconstruction; world models; diffusion models; flow matching; uncertainty estimation.
LLMs, NLP & Retrieval: transformer models; supervised fine-tuning; RLHF; RLAIF; DPO; RAG; vector databases; information retrieval; document indexing; agentic tool selection.
Finance & Quant Research: market microstructure; alpha research; time-series forecasting; risk modelling; backtesting.
Systems, Data & Deployment: Linux; Git; Docker; Apptainer; SLURM; CUDA programming; Numba; CuPy; `torch.compile`; ONNX Runtime; Weights & Biases; AWS — EC2, S3; REST APIs; backtesting frameworks; AMD Vivado.